

EV Charger Circuit Fundamentals

Category: EV Charging Infrastructure

This overview highlights the main electrical considerations when adding EV chargers to a building.

- **SUPPLY RATING:** Determine available spare capacity at the main incomer and any intermediate panels. Consider diversity,
- **CIRCUIT SIZING:** Select cable sizes and protective devices according to charger rating, installation method and ambient conditions.
- **RESIDUAL CURRENT PROTECTION:** Follow charger manufacturer guidance for RCD type (e.g., Type A with DC detection).
- **LOAD MANAGEMENT:** For multiple chargers, consider dynamic load management to prevent main breaker trips.
- **EARTHING AND BONDING:** Pay particular attention to earthing arrangements, especially in outdoor car parks.

Always cross-check with the manufacturer's installation manual and applicable national standards.